Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · Cloud-Native · AI/ML Integration · CI/CD · Scalable Systems · Financial Data Pipelines · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience delivering scalable, secure, and maintainable software systems through the full development lifecycle - design, coding, automated testing, CI/CD, deployment, and ongoing production support. At Entrata, I owned mission-critical financial data pipelines for 9 years: a federal compliance pipeline processing income-limit data for 180,000 domain records that power eligibility and payment calculations for 100K+ housing units, where an incorrect record means incorrect payments and regulatory violations. I built and maintained these systems with 99.5% uptime, 100% PHI compliance, zero security incidents, and full auditability - the reliability and accuracy standards that financial operations require. I integrate AI and ML components into production systems: AgentProbe is a generative AI evaluation platform I built end-to-end (multi-provider LLM integration, automated regression testing, CI quality gate), and LeaseI is a production AI system with automated document intelligence and human-review workflows. I leverage generative AI tools (Claude Code) daily to accelerate development velocity - writing, testing, reviewing, and deploying through agentic coding workflows. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr), TypeScript, JavaScript, PHP, SQL, PL/SQL, Bash, FastAPI, REST APIs, React.js, HTML/CSS, jQuery

AI/ML & Generative AI: AI/ML Integration, LangGraph, LangChain, RAG Pipelines, LLM APIs (Anthropic, OpenAI, Ollama), Generative AI Tools (Claude Code), Agentic AI, LLM Evaluation, Automated ML Testing, PyTorch, scikit-learn, BERT, spaCy, NLP, Prompt Engineering

Cloud-Native & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (async processing), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Automated Testing, Performance Engineering, Observability, Defect Management, PHI/HIPAA Compliance, Agile/Scrum, Linux, Git

PROJECTS

AgentProbe: AI/ML Integration Platform with CI/CD Automated Testing

AI/ML Integration · Generative AI · CI/CD · Automated Testing

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production platform integrating AI/ML components into software delivery workflows - multi-provider generative AI integration (Anthropic, OpenAI, Ollama) via clean API abstractions; CI/CD-integrated automated testing that gates quality before deployment; 5-metric evaluation engine measuring correctness, performance, and reliability of AI outputs. Built end-to-end with full documentation, automated test coverage, and continuous improvement feedback loop. Zero regressions across 1,240+ test cases. Open-source and enterprise-deployable.

LeaseI: Production AI System - Automated Processing with Human Review

AI Integration · Scalable · Secure · Full Lifecycle

Python · LangGraph · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable, secure, maintainable AI system built through the full software development lifecycle - automated document processing (OCR, section-aware extraction), AI-powered analysis with human-review escalation for high-stakes decisions, and end-to-end test coverage with CI regression gate. Designed with auditability, data governance, and compliance as first-class requirements. 93.6% precision across 1,240 test cases; zero unsupported-claim rate on production eval set.

AI on FHIR: Cloud-Native Full-Stack Application - HIPAA Compliant

Cloud-Native · Secure · API Integration · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Cloud-native full-stack application integrating AI/ML components with enterprise REST APIs - HIPAA-compliant RBAC with authentication and full data isolation; NLP-powered query translation reducing manual retrieval time ~60%; containerized with Docker for scalable deployment. Demonstrates secure, compliant, maintainable software aligned with enterprise standards.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production AI Delivery

- Designed and delivered a production AI system (Python, LangGraph, FastAPI) for an external customer end-to-end - gathering requirements from stakeholders, architecting the solution, building automated testing, and deploying to production in weeks as sole technical owner.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering; translated complex technical outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Financial data pipelines at production scale** - owned the HUD income-limit pipeline for 7 years (federal data → 180,000 domain records powering eligibility and payment calculations for 100K+ housing units); annual releases with baseline validation, reconciliation, and rollback preparation; zero compliance violations across 7yr ownership. Automated 80% of manual operational workload through Python/Bash pipelines, directly reducing manual effort and improving financial process accuracy.
- **Scalable, secure, maintainable software delivery** - AWS S3/EC2/RDS/Glue; Docker microservices (50% scalability improvement, 30% deployment time reduction); RabbitMQ async processing with retry logic and idempotency guarantees; 200+ SQL optimizations (30% performance gain); 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across 7yr; GitHub Actions CI/CD; automated testing throughout.
- **Full lifecycle delivery and collaboration** - 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules shipped end-to-end; code reviews (300+ peer reviews); internal debugging tools reducing triage time 40%; mentored 5+ engineers; leveraged Claude Code (generative AI tools) to improve development velocity and code quality.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; financial data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017